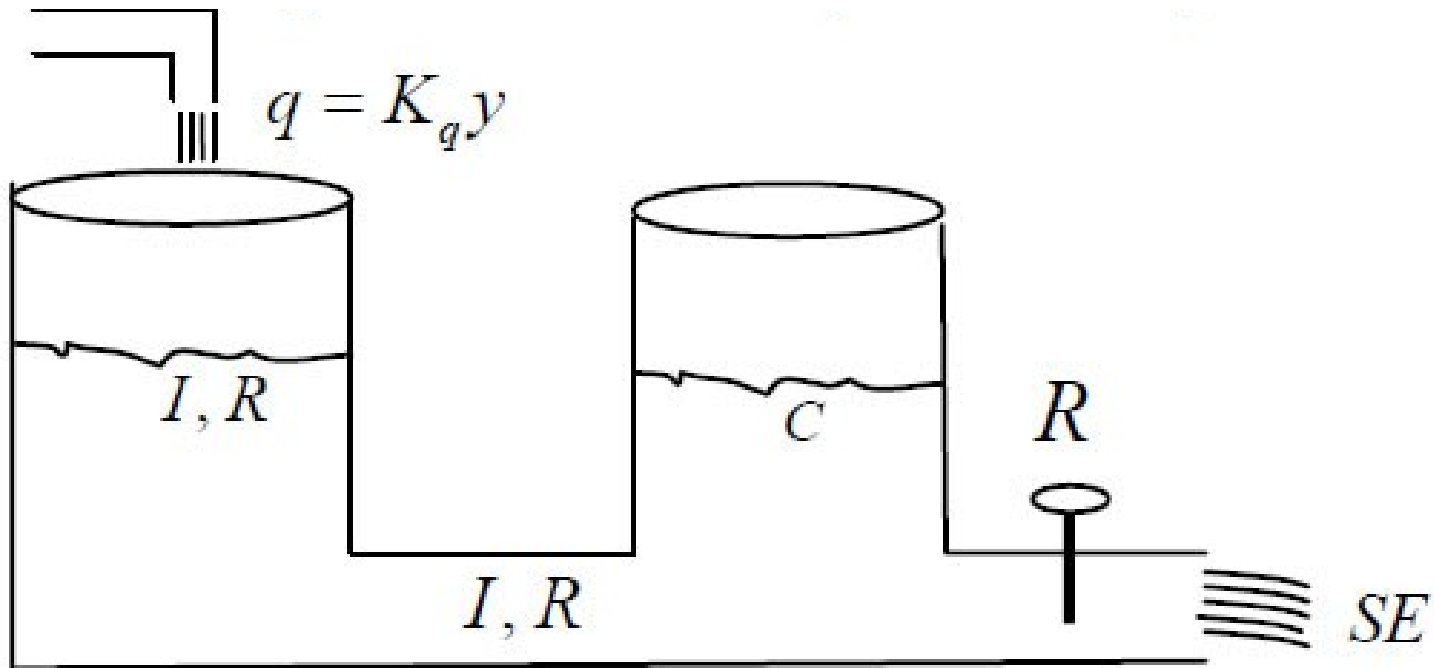


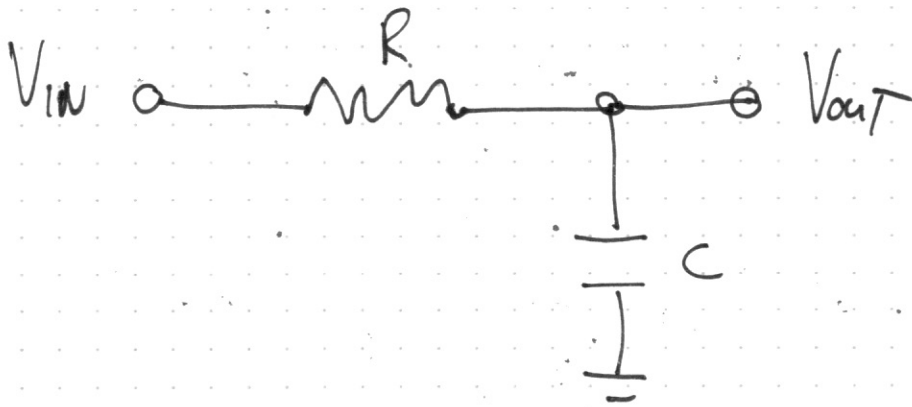
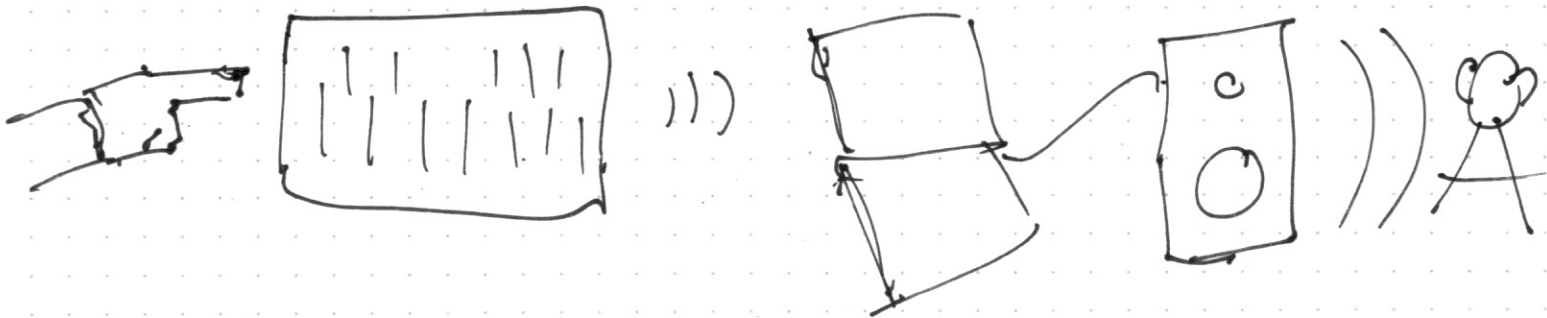
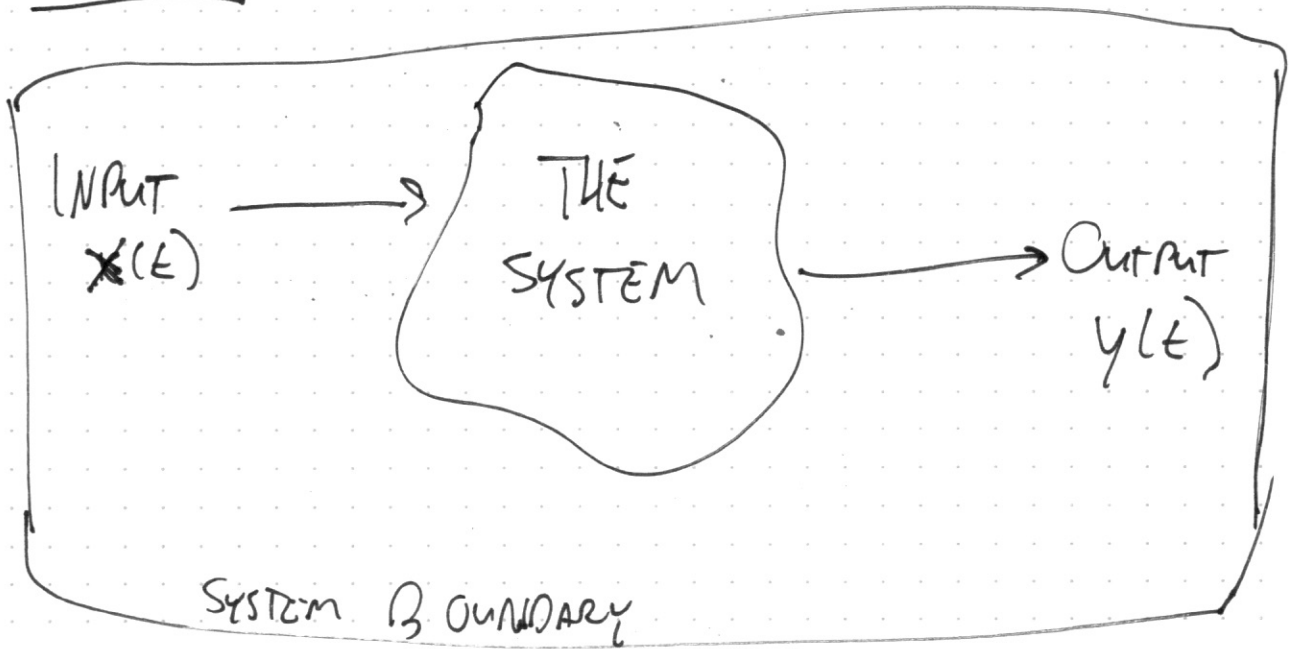


SYDE252 - W02B – LTI Systems
Dr. Matt Borland

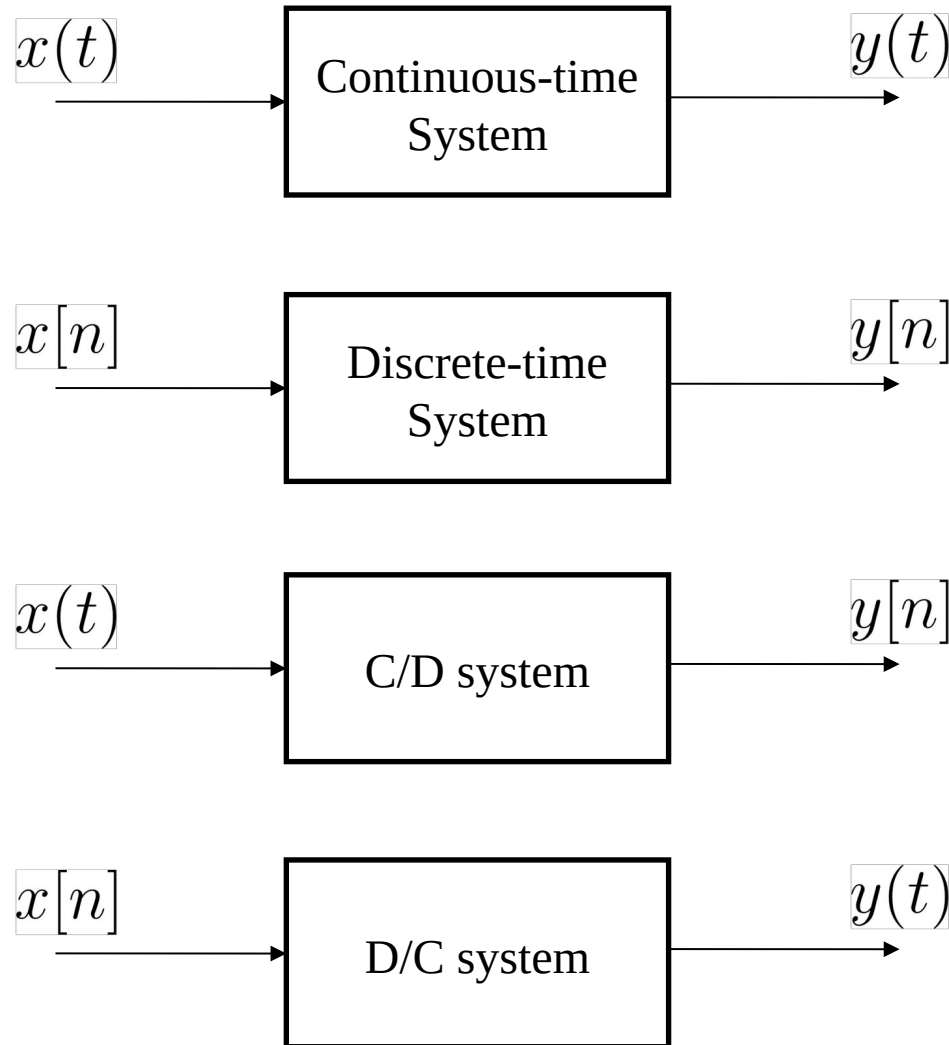
What are we doing today:

- Introduce some basic properties of systems

SYSTEMS

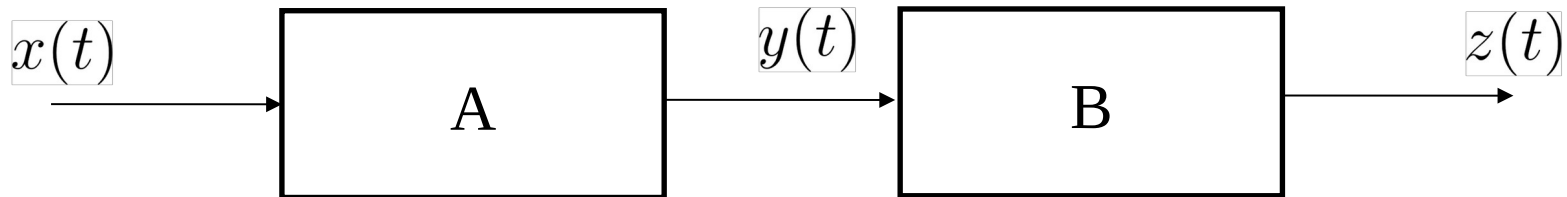


Basic concepts for systems

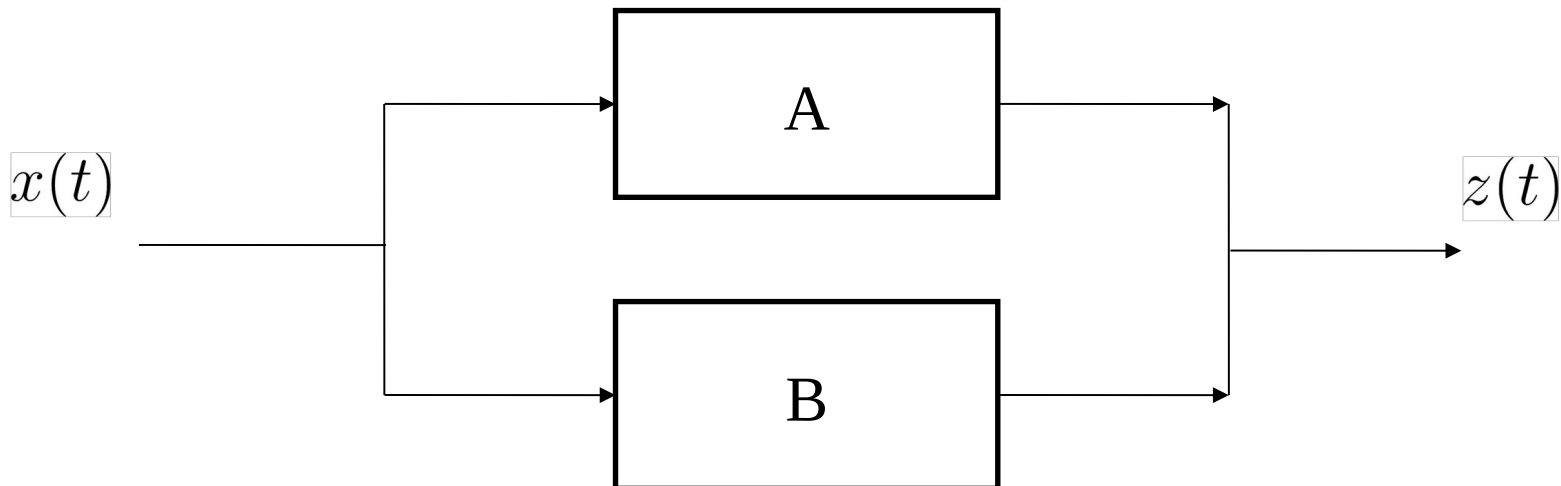


Basic concepts for systems

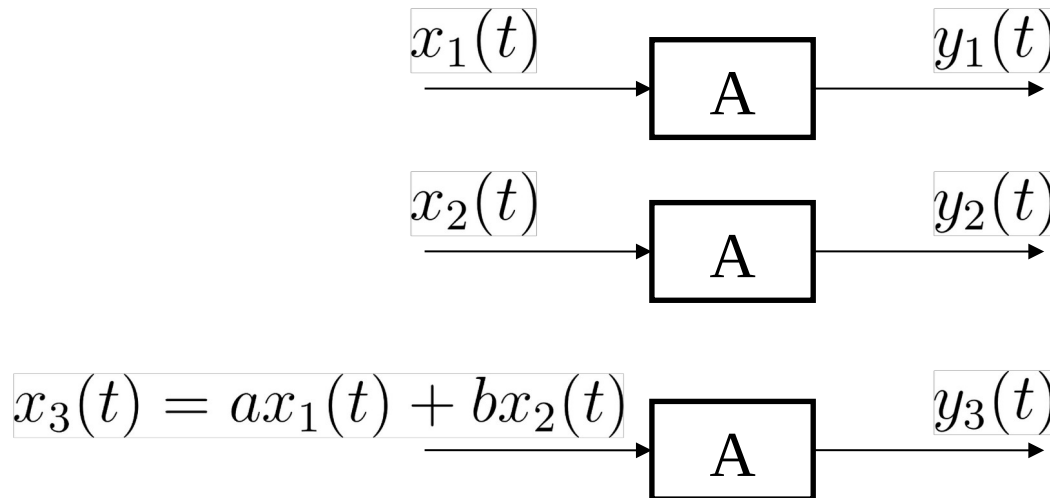
Cascading systems



Parallel systems

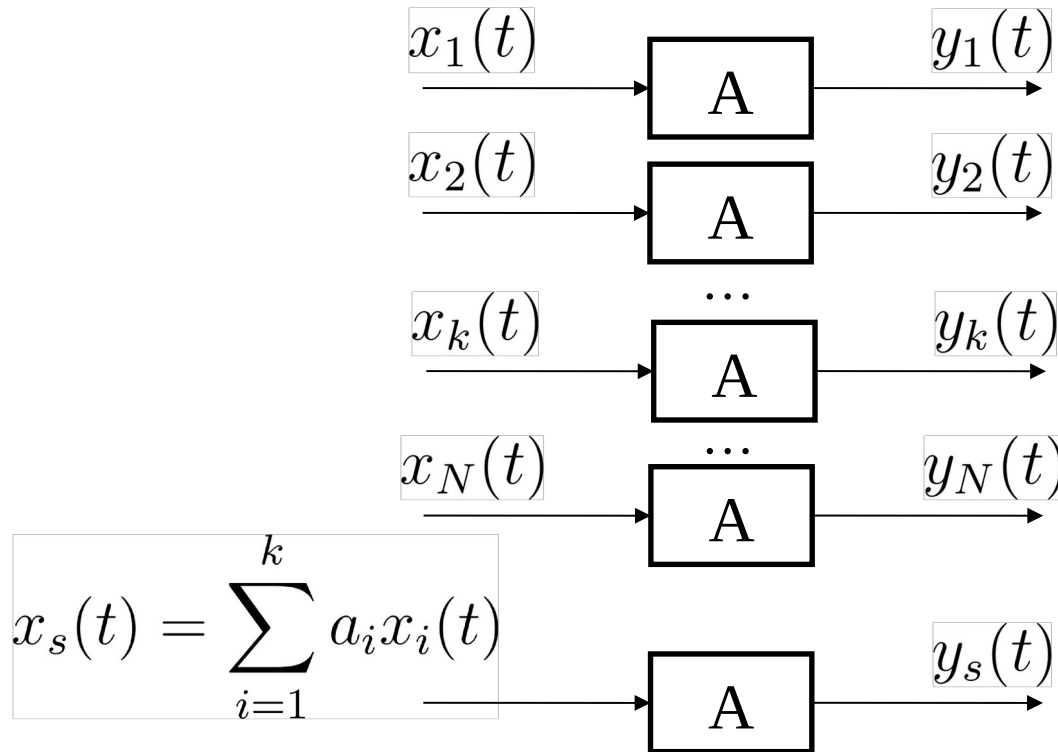


Basic concepts for systems



If and only if $y_3(t) = ay_1(t) + by_2(t)$, system A is called **Linear**.

Linearity of a System



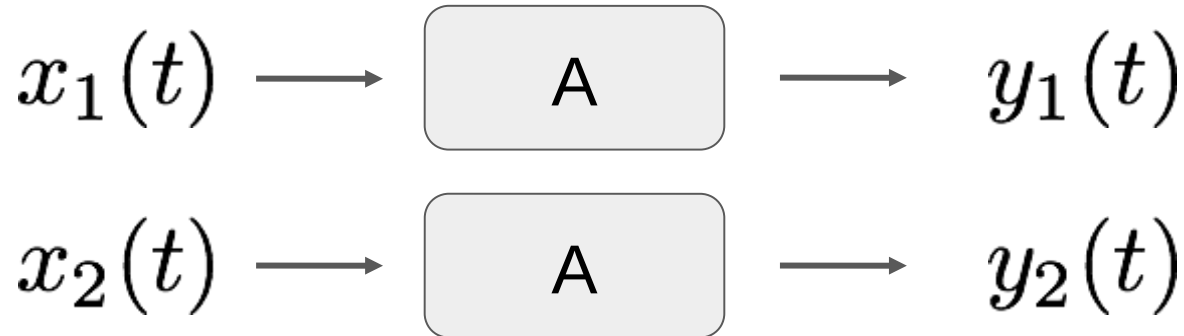
If and only if $y_s(t) = \sum_{i=1}^N a_i y_i(t)$, system A is called **Linear**.

Also called the SUPERPOSITION property!

In a bit different language:

For the input signals:

...and the system outputs:



The system is linear if:

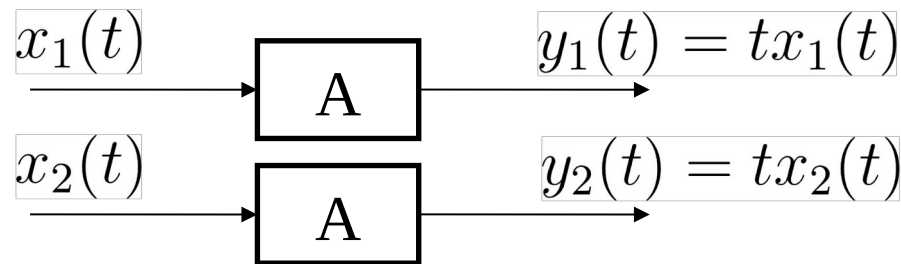
1. The response to $x_1(t) + x_2(t)$ is $y_1(t) + y_2(t)$
2. The response to $ax_1(t)$ is $ay_1(t)$, where a is any complex constant

1. Is the additivity property.
2. Is the scaling (or homogeneity) property.

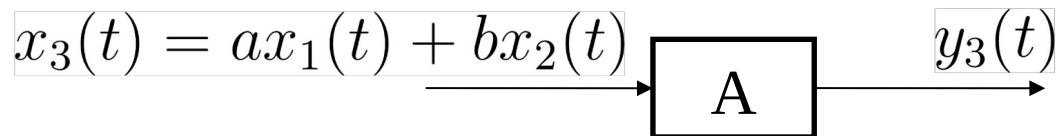
Basic concepts for systems

Linearity

Example 1: Is the system below Linear?



When we have an input:



$$y_3(t) = tx_3(t)$$

$$= t(ax_1(t) + bx_2(t))$$

$$= tax_1(t) + tbx_2(t)$$

$$= ay_1(t) + by_2(t) \quad \Rightarrow \quad \text{The system } y(t) = tx(t) \text{ is } \mathbf{Linear}.$$

LINEARITY



For
L
I
N
E
A
R
I
T
Y

Additivity

$$x_1(t) + x_2(t) \rightarrow \boxed{A} \rightarrow y_1(t) + y_2(t)$$

Homogeneity

$$a x_1(t) \rightarrow \boxed{A} \rightarrow a y_1(t)$$

Ans

$$x_1(t) \rightarrow \boxed{A} \rightarrow y_1(t) = t x_1(t)$$

$$x_2(t) \rightarrow \boxed{A} \rightarrow y_2(t) = t x_2(t)$$

$$x_3(t) = (x_1(t) + x_2(t)) \rightarrow \boxed{A} \rightarrow y_3(t) = t (x_1(t) + x_2(t))$$

$$y_1(t) + y_2(t) = t x_1(t) + t x_2(t)$$

$$= t (x_1(t) + x_2(t))$$

$$= y_3(t)$$

$\therefore$ Additivity works

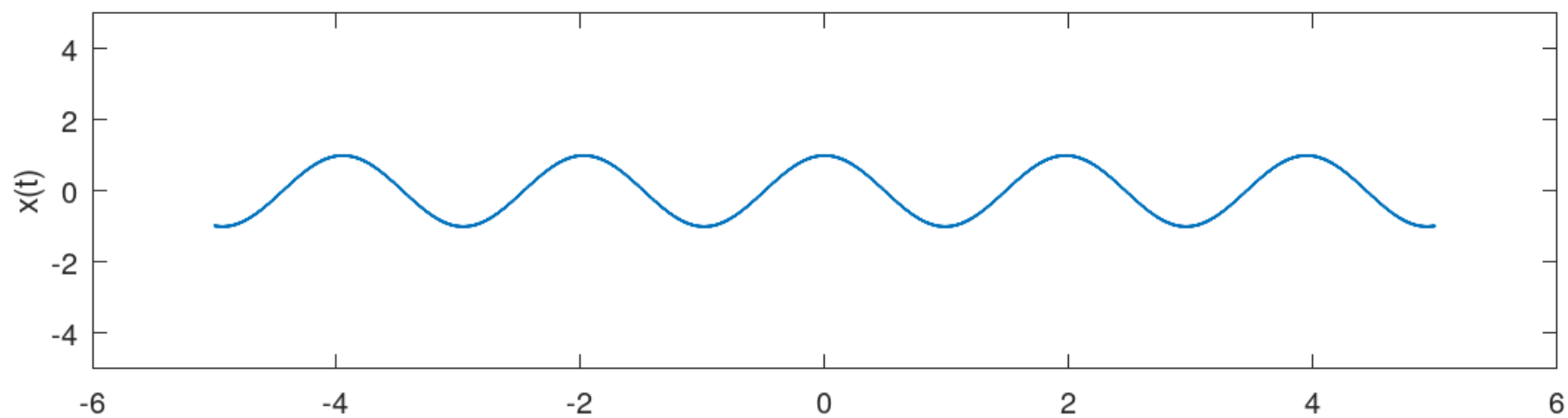
$$x_1(t) \rightarrow \boxed{A} \rightarrow y_1(t) = t x_1(t)$$

$$a x_1(t) \rightarrow \boxed{A} \rightarrow y_2(t) = t (a x_1(t))$$

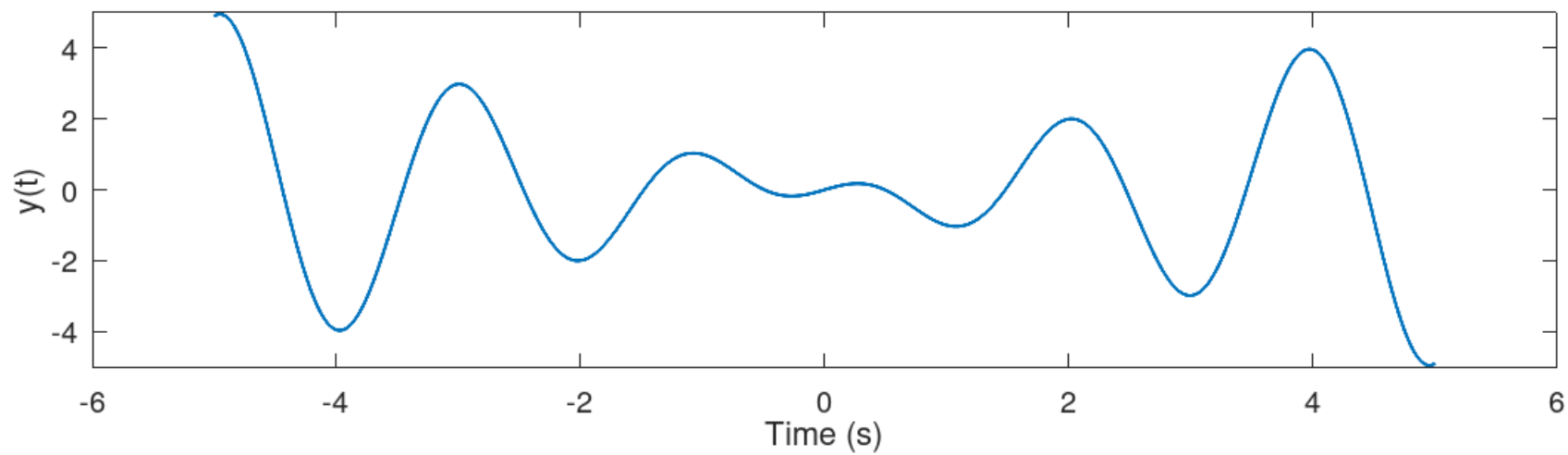
$$y_2(t) = a y_1(t)$$

$$\therefore a x_1(t) \rightarrow \boxed{A} \rightarrow a y_1(t)$$

INPUT



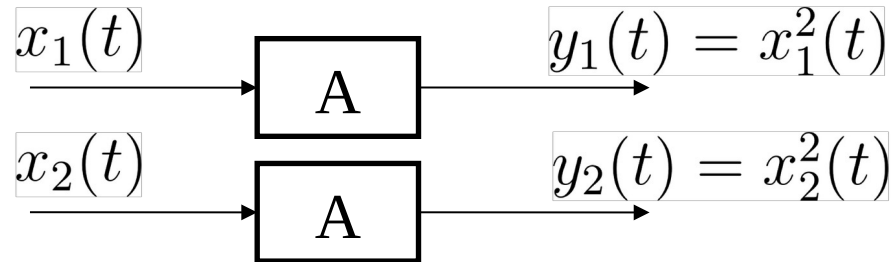
OUTPUT: $y(t) = t \cdot x(t)$



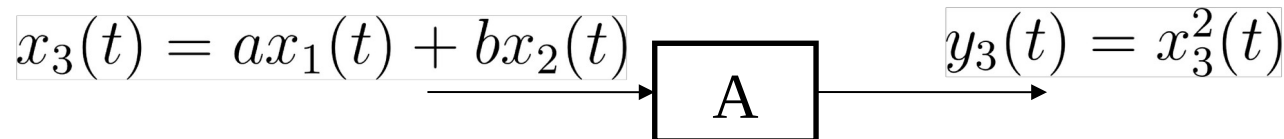
Basic concepts for systems

Linearity

Example 2: Is the system below Linear?



When we have an input:



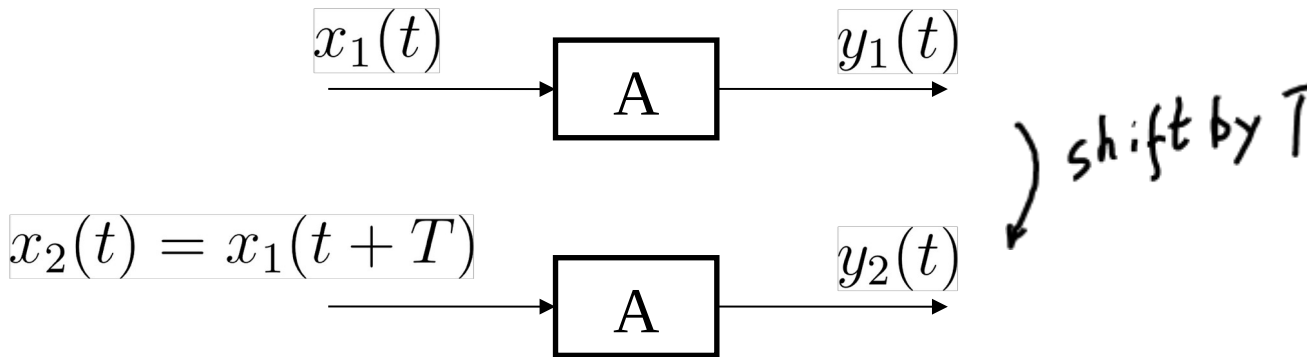
$$\begin{aligned}
 y_3(t) &= x_3^2(t) \\
 &= (ax_1(t) + bx_2(t))^2 \\
 &= a^2x_1^2(t) + b^2x_2^2(t) + 2abx_1(t)x_2(t) \\
 &\neq ay_1(t) + by_2(t) \quad \Rightarrow
 \end{aligned}$$

The system

is **NOT Linear**.

Basic concepts for systems

Time-invariance



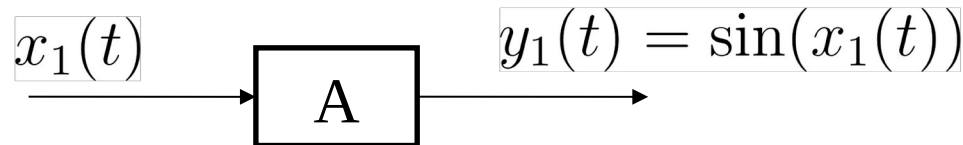
If and only if $y_2(t) = y_1(t + T), \forall t, \forall T$ system A is called **Time-invariant**.

Otherwise, the system is called time-variant.

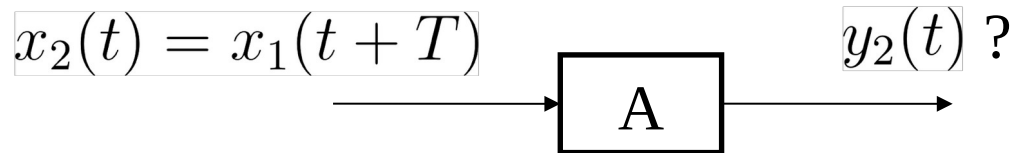
Basic concepts for systems

Time-invariance

Example 1: Is the system below time-invariant?



When we have an input:



When the input is $x_1(t + T)$ the output is:

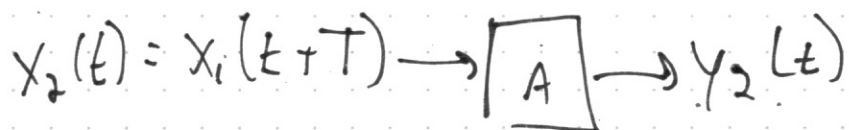
$$y_2(t) = \sin(x_2(t)) = \sin(x_1(t + T))$$

the shifted original output is:

$$y_1(t + T) = \sin(x_1(t + T))$$

Therefore $y_1(t + T) = y_2(t) \Rightarrow$ The system is **Time-invariant**

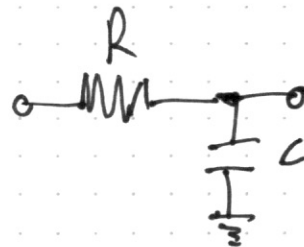
TIME INVARIANCE.



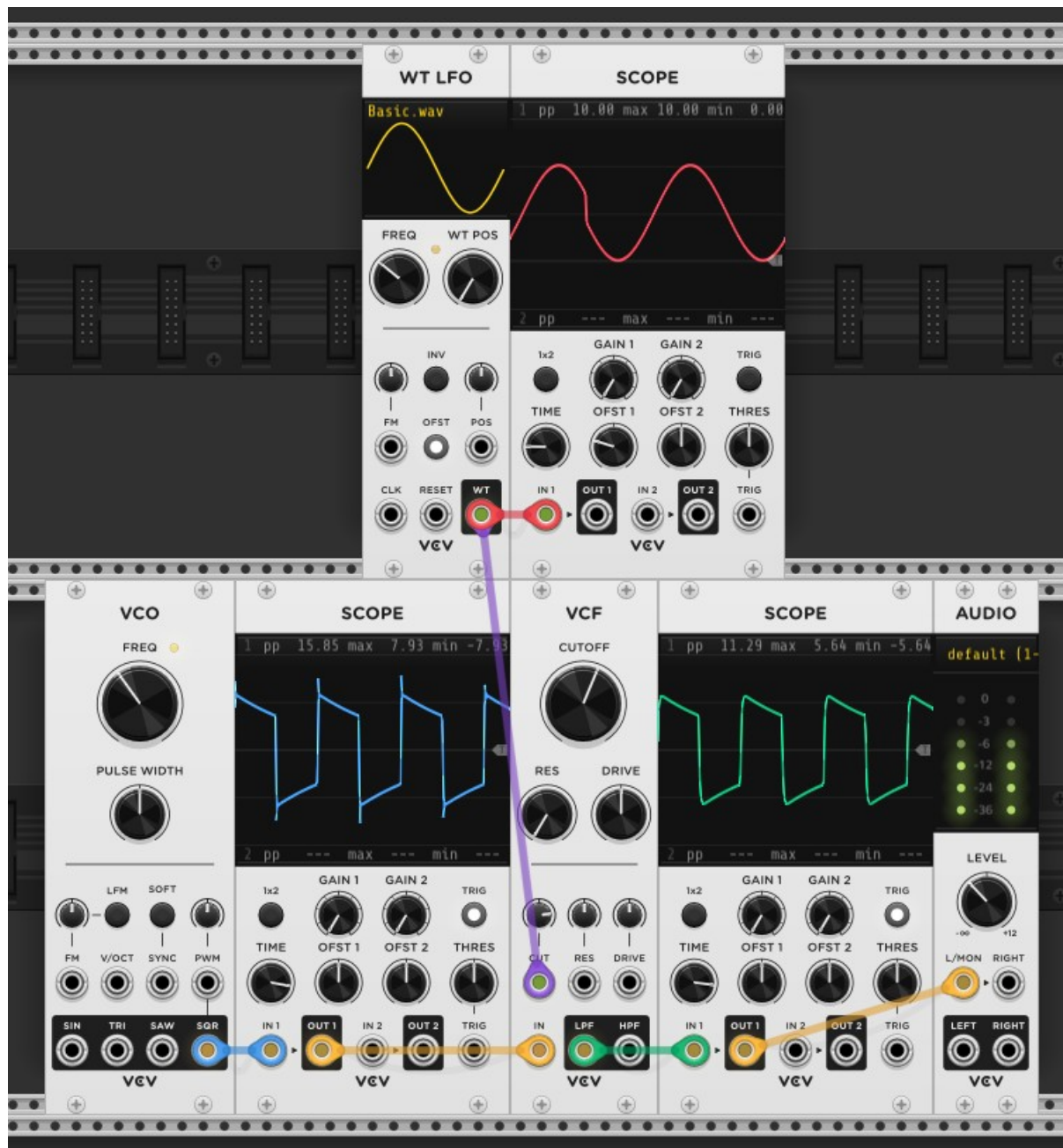
T.I. IF $y_2(t) = y_1(t+T)$

- A SHIFT IN THE INPUT RESULTS IN ONLY THE SAME TIME SHIFT IN THE OUTPUT

→ THE TIME AT WHICH YOU PUT-IN ~~TO~~ THE INPUT DOESN'T CHANGE THE OUTPUT RESPONSE, OTHER THAN WHEN THE OUTPUT OCCURS



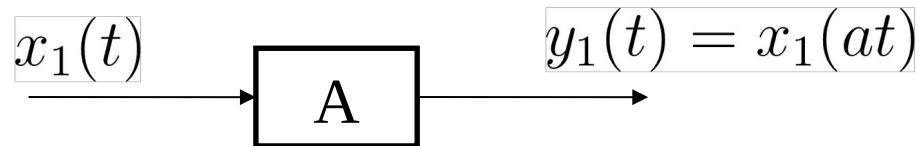
WHAT CAN I DO TO MAKE A LPF TIME VARIANT?



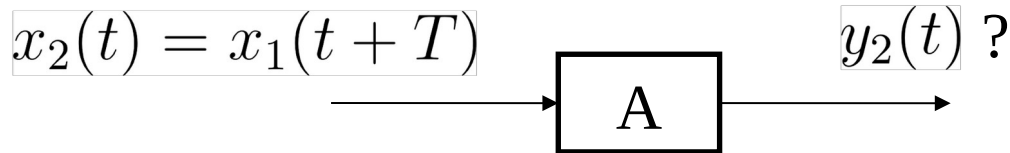
Basic concepts for systems

Time-invariance

Example 2: Is the system below time-invariant?



When we have an input:



When the input is $x_1(t + T)$
the output is:

$$y_2(t) = x_2(at) = x_1(at + T)$$

the shifted output
original is:

$$y_1(t + T) = x_1(a(t + T))$$

Therefore $y_1(t + T) \neq y_2(t) \Rightarrow$ The system is **Time-variant**